# Reflection Journal Report for Lab 02

**Prepared for Prof. Vishwantha Rao, HCC ITAI 1371 Machine Learning**
**Prepared by FA26_ML_6171_16347 3_ITAI1371 Group**
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The purpose of this report is for each student to reflect on their learning journey for ITAI 1371 Machine Learning (ML) Lab 02. This Reflection Journal Report consists of separate parts for each student in the group. Each student in the group was responsible for their own content and report formatting in each of the sections below.

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This reflection journal entry is for Lab 02 Google Collab and was prepared by Xiomara Torres.

## Overview

The Module 2 lab introduces several tools used in machine learning. Some of those tools are Google collab, Pandas, NumPy, matplotlib, and the Iris flower dataset which is what we used and the data we'll be reflecting on. It also introduced GitHub and how completed projects can be documented as stored portfolio pieces.

## Purpose

The purpose of this lab is to develop fundamental skills using common machine learning tools. This reflection will discuss our experience working towards this objective, challenges encountered, and how we will apply these skills in future assignments.

## Description of Experience:

Before completing this lab, I had never used Google Collab before, so there was a learning curve. Terms such as "run" "push" "repository" and "cells" were unfamiliar to me. Although the assignment was designed for beginners, the language used made instructions appear more complicated than they really were. One instruction that confused me initially is "run all cells". At first, I believed I needed to copy code from google collab and paste it onto Python on my computer and run it there. After some googling, I realized I just had to click play on the cell. Once I understood the instructions, I ran the code and observed how the notebook displayed the information from the Iris dataset, which appeared immediately.

## Personal Reflections:

At the beginning of this assignment, I felt completely confused and slightly irritated. My frustration did not come from the work being too difficult, but from feeling as though I was stepping into a new world and seeing terminology I had never encountered before. I did not immediately understand what the assignment was expecting me to do. Once the instructions were simplified, I realized that I had been overcomplicating the entire process. This realization helped me feel more confident and reminded me that confusion is a normal part of learning something new and does not mean I am incapable or unfit.

## Discussion of Improvements and Learning

This experience has taught me to be more patient with myself when learning unfamiliar technology. Throughout the lab I developed basic skills in navigating Google Collab, running notebook cells, and understanding how code works with data. I also became more familiar with the roles of NumPy, pandas, and matplotlib in machine learning.

## Conclusion:

Overall, this lab taught me how to navigate introductory machine learning tools. I understand the importance of clean, clear code, and organized data. I also learned that technical assignments are more manageable when I break them down into smaller parts, work through them patiently, and ask for clarifications when necessary.

This reflection journal entry is for Lab 02 Google Collab and was prepared by Eric Roberson.

## I. Understanding Insights & New Breakthroughs

Everything about Machine Learning (ML) iOS new to me, which means everything new that I learn will be a breakthrough. "Google Colaboratory (Colab) is Google's free cloud-based version of Jupyter notebooks, designed specifically for machine learning and data science" (ITAI 1371 Module 2 Essential Development Environments for ML PowerPoint, Slide 7). Google Colab is cloud-based, runs in a Web browser, and provides GPUs and TPUs for free. It is good for beginners. I was familiar with the acronym graphical processing units (GPUs) prior to the lab. I had to look up tensor processing unit (TPU) to understand what it meant. Libraries are pre-installed and available to for use through the product, which makes Google Collab perfect for beginners. Files are automatically saved to Google Drive; however, I'm unsure how large files would grow in a real-world environment. So, it's likely the free version of Google Drive will not be sufficient for production work. Another breakthrough for me is that it is a co-lab, and not a collaboration between Google and a third-party. This could be confusing for someone with a non-technical background.

## II. Analyzing the Problem / Problem Solving Strategies

I learned that the integration of Colab with GitHub can be flaky. Although the authentication to connect the two tools appeared on first run, the file would not copy. In other words, the File > Copy to GitHub feature did not work. So, I had to manually upload my .IPYNB file from Google Drive to GitHub.

## III. Connecting New & Prior Knowledge

I already knew about libraries and datasets from my Intro to AI course and from my day job. What I learned in this lab was how easy it is to import a dataset

## IV.

I'm not sure this lab is an accurate representation of how Colab will work in the real world with real data, but it was good practice using the beginner Iris dataset. There are over 300 species of Iris, and I would like to know how do other species compare to these samples? It would be interesting to learn how to expand the dataset to broaden the samples to include more species. I would expect in the business world that datasets grow and change frequently.  Top of mind would be monthly sales info since books close at the end of every month, or key performance indicators (KPIs) and keys risk indicators (KRI) in the IT field.

## V. Changing Approach & Critical Thinking

The biggest change in my thinking is how rapidly datasets will grow when working with real data in the real world.  Business data is constantly changing, and with real-world business data, the administration of keeping the datasets fresh would be an important job.  I would think that the process could be automated to expand datasets in the business world.

## VI. Markdown from the Lab Exercise

My Observations About the Iris Dataset are as follows:

**Dataset Overview**

Number of samples: 150

Number of features: 4

Number of classes: 3

**Key Findings from the Visualization**

The Setosa species forms a distinct cluster above the other two species, Versicolor and Virginica. There is a clear gap between Setosa and the other species.

Setosa clearly has a greater sepal width compared to the other two species, which have a greater sepal length.

Identical number of samples (n=50) were collected for all three species according to the bar chart.

**Questions for Further Investigation**

There are over 300 species of Iris, and I would like to know how do other species compare to these samples? It would be interesting to learn how to expand the dataset to broaden the samples to include more species.

I'd also like to know how to compare petal widths and leghts to the sepal widths and lengths. So, how do you create a visual 3D matrix?

## **Reflection**

I genuinely like the cell feature of Colab and how the lab is divided into 16 distinct cells. I can see how this could be useful for modularization with programming.

This reflection journal entry is for Lab 02 Google Collab and was prepared by Irene "Alex" Ramos.

**Reflective Journal for Irene(Alex) Ramos**

## 1. Introduction

### Brief Overview

This lab gave me a much better understanding of what actually goes into working with machine learning. Before doing the lab, I mostly thought about machine learning as using data to train a computer to recognize patterns or make predictions. I did not really think about all the tools and steps that happen before you even get to that point. The lab introduced tools like Google Colab, Python libraries, data visualization, and GitHub. I also learned quite a bit about working with a group online, which ended up being one of the more challenging parts for me.

### Purpose

The purpose of this reflection is to discuss what I actually learned from the lab instead of just going over what I completed. The biggest things I took away from it were learning how visualization can help me understand data, becoming more comfortable troubleshooting when something does not work, and realizing that online group projects require a lot more communication and planning than I expected.

## 2. Description of Experience or Topic

### Background Information

Going into the lab, I had a basic understanding of machine learning, but I was still learning how the different tools fit together. One thing that became clearer to me is that you cannot just throw data into a machine learning model and expect useful results. You have to understand what you are working with first.

The Iris dataset was a good example of this. It contained 150 samples, four different measurements, and three species of Iris flowers. At first, it was basically just a bunch of numbers to me. Once I started working with the data and visualizing it, I could actually see what those numbers were telling me.

### Specific Details

One of the parts that stood out to me was creating the scatter plot comparing sepal length and sepal width. Looking at the numbers in the DataFrame was useful, but the graph made the differences between the species much easier for me to understand.

I also ran into a problem trying to get Google Colab to work with GitHub. The authentication appeared to work, but the **File > Copy to GitHub** feature would not actually copy my notebook. I eventually stopped trying to make that feature work and manually uploaded my `.ipynb` file from Google Drive to GitHub instead. This was something I documented while completing the original lab.

Another challenge was working with my group completely online. I had to get more comfortable sending group messages, communicating what was going on, and making sure everyone knew what needed to happen. I learned pretty quickly that online group work does not just come together on its own.

## 3. Personal Reflection

### Thoughts and Feelings

I actually liked working in Google Colab more than I expected. I genuinely like the way everything is divided into cells because I can work on one part, run it, and see what happens before moving on. I can see how that will be useful when we start doing more complicated programming.

The GitHub problem was frustrating at first because I thought I was doing something wrong. Eventually, though, I realized I did not necessarily need to fix the feature itself. I just needed to find another way to get my notebook where it needed to go.

Working with my group was probably a bigger learning experience than I expected. Communicating online is different from sitting next to somebody and working on an assignment together. I had to actually reach out, send messages, wait for responses, and make sure everyone was on the same page. That took more time than I originally planned for.

## Analysis and Interpretation

One of the biggest things that changed for me was how I think about data visualization. Before this lab, I mostly thought graphs were something you made to show other people your results. Now I understand that creating a visualization can actually help **me** figure out what is happening in the data.

For example, I noticed that Setosa formed a pretty clear cluster away from Versicolor and Virginica. Setosa also tended to have a greater sepal width, while the other two species generally had greater sepal lengths. I understood this much better after seeing it visually than I did from just looking at the numbers.

It also made me start asking more questions. I wanted to know what would happen if I compared petal length and petal width instead. I also wondered if there was a way to visualize several of those measurements at the same time. Those were questions I actually started thinking about while completing the lab.

The group project taught me something completely different. I realized that knowing how to do my own part is not enough when other people are involved. My schedule affects them, and their schedules affect me. If I wait too long to start, there is not much room for someone to take a while to respond or for the group to fix a problem.

That is probably one of the biggest lessons I am taking away from this assignment: **I need to start group projects earlier.**

## Connections to Theoretical Knowledge

The lab helped me connect the idea of machine learning with actually finding patterns in data. For example, we calculated the mean and standard deviation for sepal length. My results were a mean of about 5.84 cm and a standard deviation of about 0.83 cm. Those statistics tell me something about the dataset, but the visualizations helped me understand the relationships in a different way.

I also started to understand why the different tools are useful together. Colab gives me a place to write and test code, Python libraries help me work with the data, visualizations help me understand it, and GitHub gives me a place to keep and share the project. The lab specifically explains that GitHub can be used for version control, collaboration, a portfolio, and backup.

After working with my group, the collaboration part makes a lot more sense to me. Having tools that allow people to share work is useful, but those tools do not replace actually communicating with your team.

### Critical Thinking

I think the hands-on parts of the lab worked the best for me. Running code and immediately seeing the output helped me understand what the code was actually doing instead of just reading about it.

Strangely, I also think the GitHub problem taught me a lot. My first reaction was basically, "Why isn't this working?" Eventually, I changed the question to, "What am I actually trying to accomplish?" I needed my notebook on GitHub. Manually uploading it accomplished that, even though it was not the method I originally expected to use.

The main thing I would do differently is start the group portion sooner. I underestimated how much time online communication can take. You cannot send a group message and assume everybody will see it and respond immediately. Starting sooner would have given us more time to communicate, figure out responsibilities, and deal with problems without feeling rushed.

## 4. Discussion of Improvements and Learning

### Personal Growth

I think this lab taught me more than I expected it to. Obviously, I learned more about machine learning tools, but I also learned more about how I personally work through problems.

I became more comfortable trying another solution when the first one failed. I also had to get outside of my comfort zone a little with the group portion and communicate more. Sending group messages and keeping up with everyone might seem simple, but it is something I had to actively work on during this assignment.

## Skills Developed

One skill I developed was being able to look at data in different ways instead of only looking at raw numbers. I also became more comfortable using Colab, working with Python libraries, interpreting graphs, and troubleshooting problems.

The other skill I developed was online collaboration. I learned that I have to communicate earlier and not wait until I need something immediately before reaching out to my group.

Probably the most useful lesson was realizing that both programming and group work require problem-solving. When the Colab-to-GitHub method did not work, I found another method. When working online made communication harder, I had to start communicating more actively. In both situations, I had to adjust instead of expecting everything to work exactly the way I planned.

## Future Application

The biggest change I am going to make in future assignments is starting group projects earlier. I know now that I need to give myself extra time when other people's schedules are involved. I want to contact my group earlier, figure out how we are going to communicate, and make sure everyone knows what we are doing before we get close to the deadline.

I also want to experiment more during future labs. Instead of only running the code that is required, I want to change things and see what happens. The next lab is supposed to introduce different types of machine learning and building a simple classifier. I am interested in seeing whether the patterns I could see in the Iris graph are similar to the patterns the classifier will use.

# 5. Conclusion

## Summary

Overall, this lab taught me a lot about machine learning, but some of the most important lessons were not just about Python. I learned that visualizing data can help me understand it instead of simply being a way to present results. I learned that when technology does not work the way I expect, sometimes the best solution is finding another way to accomplish the same goal. I also learned that working with a group online takes real communication and planning.

## Final Thoughts

The biggest thing I will do differently going forward is **start group projects earlier**. This lab made me realize how easy it is to underestimate the amount of time it takes to communicate online. People have different schedules, messages are not always answered immediately, and problems can come up that nobody planned for.

I also want to keep the curiosity I started developing during this lab. Instead of just asking whether my code worked, I found myself asking why the data looked the way it did and what would happen if I compared different features. That is probably the part of the lab that taught me the most. I am starting to understand that learning machine learning is not just about getting the code to run. It is about understanding what the code and data are actually telling me.

This reflection journal entry is for Lab 02 Google Collab and was prepared by Yuanyuan Kou.

# Student: Yuanyuan Kou Group3

# Course: ITAI 1371

## 1. Beginning the Lab

Before this lab, I did not have a clear understanding of what a notebook was. I first saw the .ipynb file as an unfamiliar type of document and was unsure where to begin. After opening it in Google Colab, I realized that a notebook is like an interactive workbook that combines explanations, Python code, results, and graphs. This understanding made the assignment feel more manageable because I could work through one cell at a time instead of treating the whole file as one large program.

## 2. Understanding the Roles of the Tools

The most important insight for me was seeing how pandas, NumPy, and Matplotlib have different but connected roles. Pandas organized the Iris data into a DataFrame that looked familiar because of my experience with Excel. However, I could see that a DataFrame is more powerful because Python can select, group, count, and summarize data automatically. NumPy handled the numerical calculations, such as the mean and standard deviation of sepal length. Matplotlib then changed numerical information into visual information through a scatter plot and a bar chart. I began to understand that data analysis is a workflow: first organize the data, then calculate useful information, and finally visualize patterns.

## 3. Learning from the Iris Dataset

The visualizations helped me understand why data exploration is necessary before building a machine learning model. The bar chart showed that the dataset is balanced because all three flower species have 50 samples. In the scatter plot, setosa appeared more separate, while versicolor and virginica overlapped when sepal length and sepal width were compared. This made me question whether petal measurements would separate the species more clearly. I also learned that features are the measurements used for prediction and that the species column represents the target class that a future model would try to predict.

## 4. Challenges and Problem-Solving

One challenge was understanding why the cells must be run in order. At first, clicking a Run button seemed like a simple action, but I learned that later cells depend on variables created earlier. For example, the DataFrame named df must exist before it can be grouped, summarized, or plotted. This helped me see that an error message may not mean the code itself is wrong; it may mean that an earlier step was skipped. Running all cells from the beginning became an important way to test whether the notebook was reproducible.

## 5. How My Thinking Changed

This lab changed my view of machine learning because I had expected the main task to be training a model. Instead, I learned that understanding and preparing data comes first. A model can only learn from the information it receives, so checking the structure, balance, measurements, and visual patterns of a dataset is an essential part of the process. I still need more practice reading Python syntax, but I now understand the purpose of the main tools and feel more confident approaching a notebook systematically.

## 6. Next Steps

My next goal is to become more comfortable changing small parts of the code and predicting what the output will be. I would like to compare different pairs of Iris features and learn which features are most useful for classification. I also want to practice reading error messages instead of feeling discouraged by them. This lab gave me a foundation for the next module, where the data exploration process will connect to an actual machine learning classifier.

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This concludes the reflection journal portion of ITAI 1371 ML Lab 02, which was uploaded to GitHub on 30-Aug-2026. Each student was responsible for the content and formatting of their individual sections of this report.

Thank you for the opportunity to reflect on our learning for Lab 02.